

**EBN111 SEMESTER TOETS/TEST 1 (12 MAART/MARCH 2015)**  
**ANTWOORDBLAD / ANSWER SHEET**

|                         |      |                                    |     |  |     |  |  |  |
|-------------------------|------|------------------------------------|-----|--|-----|--|--|--|
| Van / Surname:          | Memo | Studentenommer:<br>Student number: |     |  |     |  |  |  |
| Voorletters / Initials: |      | Taal / Language:                   | Afr |  | Eng |  |  |  |

|    |  |    |  |      |     |
|----|--|----|--|------|-----|
| Q1 |  | Q2 |  | Tot: | /45 |
|----|--|----|--|------|-----|

| VRAAG 1 – QUESTION 1 (22) |                                                          |                    |                                                          |
|---------------------------|----------------------------------------------------------|--------------------|----------------------------------------------------------|
| 1.1                       | a) $n = 1.25 \times 10^{19} \text{ e} \checkmark$        |                    | b) Cost = 0.4375 Rands                                   |
| 1.2                       | $Q = 45 \text{ C} \checkmark \checkmark$                 |                    |                                                          |
| 1.3                       | $P = 280 \text{ W} \checkmark \checkmark$                |                    |                                                          |
| 1.4                       | a) $P_{(0.6I)} = 7.2 \text{ W} \checkmark \checkmark$    |                    | b) $P_{(RI)} = 30 \text{ W} \checkmark \checkmark$       |
| 1.5                       | $n = 5 \checkmark$                                       | $b = 7 \checkmark$ | $l = 3 \checkmark$                                       |
| 1.6                       | $i = 5.33 \text{ A} \checkmark \checkmark \frac{16}{3}$  |                    | $V = 10.67 \text{ V} \checkmark \checkmark \frac{32}{3}$ |
| 1.7                       | $I_o = 1.25 \text{ A} \checkmark \checkmark \frac{5}{4}$ |                    | $V_o = 2.5 \text{ V} \checkmark \checkmark \frac{5}{2}$  |
| 1.8                       | a) $R_{eq} = 58 \Omega \checkmark \checkmark$            |                    | b) $R_{eq} = 20 \Omega \checkmark \checkmark$            |

| VRAAG 2 – QUESTION 2 (23) |                                                                                                                                                                                                                    |                     |                                                |                     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------|---------------------|
| 2.1                       | $a = 12 \checkmark$                                                                                                                                                                                                | $b = -6 \checkmark$ | $c = 9 \checkmark$                             | $d = 0 \checkmark$  |
| 2.2                       | $V_a = 22 \text{ V} \checkmark \checkmark$                                                                                                                                                                         |                     | $V_b = 10 \text{ V} \checkmark \checkmark$     |                     |
| 2.3                       | $e = -18 \checkmark$                                                                                                                                                                                               | $f = 5 \checkmark$  | $g = -20 \checkmark$                           | $h = 40 \checkmark$ |
| 2.4                       | $\begin{bmatrix} 100 & -20 & 0 \\ -20 & 100 & -50 \\ 0 & -50 & 100 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 18 \\ -18 \\ -3 \end{bmatrix} \checkmark \checkmark \checkmark$ |                     |                                                |                     |
| 2.5                       | $i_2 = -3 \text{ A} \checkmark \checkmark$                                                                                                                                                                         |                     | $V_3 = 1 \text{ V} \checkmark \checkmark$      |                     |
| 2.6                       | $I_C = 1.59 \text{ mA} \checkmark \checkmark$                                                                                                                                                                      |                     | $I_B = 22.6 \mu\text{A} \checkmark \checkmark$ |                     |